A SMALL THEME FOR MODERN SLIDES

Cookie Beamer Theme

A LuaLaTeX Beamer theme with modern visuals

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ROADMAP

Agenda

TOC

1
Design system

1 A theme for modern talks

AWESOME-STYLE FRAME

Number rails, angled title art, and section slides are built in.

METROPOLIS DEFAULTS

Progress bars, block styles, and fonts are set from theme options.

LUALATEX SETUP

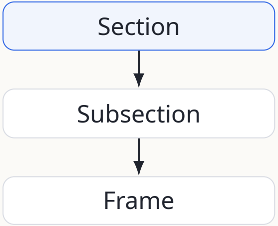
Uses installed system fonts, Unicode math, and no private package paths.

single .sty LuaLaTeX old-deck friendly

Navigation stays visible

Long titles and subtitles fit the header

- The rail shows the current section, or section.subsection when needed.
- The footer keeps author, title, section, and frame count out of the slide body.
- Section pages mark turns in the talk without adding clutter to every frame.



1 Lists and local text

Nested bullets, numbered steps, and Polish characters

- State the main point first.
  - Add the detail that changes the argument.
    - Keep the third level short.
- Return to the main thread.

1. Define the object.
2. Prove the useful property.
3. Use it in the example.

Tutaj jest krótki tekst po polsku z polskimi znakami: zażółć gęślą jaźń.

The quick, brown fox jumps over a lazy dog.

1.1

SECTION 1 / DESIGN SYSTEM

Components

1.1 Blocks and emphasis

A plain block

Use this for assumptions, notation, or a short claim.

An alert block

Use this for a warning, failed case, or decision point.

An example block

Use this for computations, test cases, or constructions.

Standard syntax

Cookie uses Beamer's normal `\begin{block}{Title}` form.

1.1 Definitions, theorems, and proofs

Definition (Divisibility)

For integers a, b, write $a \mid b$ exactly when $\exists c \in \mathbb{Z}$ such that $ac = b$.

Theorem

Every integer divides zero: $\forall a \in \mathbb{Z},\ a \mid 0.$

Proof.

Choose $c = 0$. Then $a \cdot c = a \cdot 0 = 0$ for every integer a.

1.1 Mathematics and numerals

LINING

1234567890

OLDSTYLE

1234567890

ACCENTS

$\hat{x},\ \check{x},\ \tilde{a},\ \bar{a},\ \dot{y},\ \ddot{y}$

DIFFERENTIALS

$$\iiint f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

$$\frac{1}{1+\frac{1}{2+\frac{1}{3+x}}}+\frac{1}{1+\frac{1}{2+\frac{1}{3+x}}}$$

$$F:\begin{vmatrix} F''_{xx} & F''_{xy} & F'_x \\ F''_{yx} & F''_{yy} & F'_y \\ F'_x & F'_y & 0 \end{vmatrix}=0$$

$$\iint\limits_{\mathbf{x}\in\mathbb{R}^2}\langle\mathbf{x},\mathbf{y}\rangle\,\mathrm{d}\mathbf{x}$$

$$\overline{\overline{a\alpha}^2+\underline{b\beta}+\overline{\overline{d\delta}}}$$

$$]0,1[+\lceil x\rceil-\langle x,y\rangle$$

$$\begin{aligned} e^x \approx{}& 1+x+x^2/2! \\ &+x^3/3!+x^4/4! \end{aligned}$$

$$\binom{n+1}{k}=\binom{n}{k}+\binom{n}{k-1}$$

$$\begin{aligned} p(R,\phi) \sim{}& \int_{-\infty}^{\infty}\frac{\tilde{W}_n(\gamma)\exp\left[\imath R/a\left(\sqrt{k^2a^2-\gamma^2}\cos\phi\right)\right]}{(k^2a^2-\gamma^2)^{3/4}} \\ &\times\frac{1}{H'^{(1)}_{nn}\left(\sqrt{k^2a^2-\gamma^2}\right)}\,d\gamma. \end{aligned}$$

1.1 Code is part of the slide

Scalar C

```c
void square4(float *x) {
  for (int i = 0; i < 4; ++i) {
    x[i] = x[i] * x[i];
  }
}
```

What the theme sets

- Noto Sans Mono when it is installed
- Line wrapping and padding inside the box
- Colors for keywords, strings, and comments

listings no shell escape

1.1 Longer code listing

The title wraps without changing the code box

Sleep sort sketch

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main(int argc, char **argv) {
  while (--argc > 1 && !fork())
    sleep(atoi(argv[argc]));

  puts(argv[argc]);
  return 0;
}
```

1.1 Projected gradient descent

Line numbers and comments stay readable

Algorithm 1: Projected gradient descent

Input: Objective f, feasible set C, initial point x_0, step sizes η_t

Output: Approximate minimizer x_t

for $t = 0, 1, \ldots, T-1$ **do**
  $g_t \leftarrow \nabla f(x_t)$; *// gradient at the current iterate*
  $y_{t+1} \leftarrow x_t - \eta_t g_t$;
  $x_{t+1} \leftarrow \Pi_C(y_{t+1})$; *// project back onto C*
  if $\|x_{t+1} - x_t\| < \varepsilon$ **then**
    return x_{t+1};

return x_T;

1.1 RealBoost sketch

A compact algorithm with math updates

Algorithm 2: RealBoost training loop

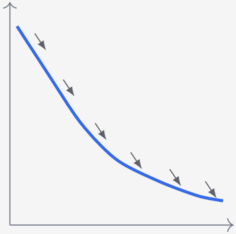
Input: Training data $(x_i, y_i)_{i=1}^{m}$, rounds T

Output: Classifier F

```
Initialize weights w_i ← 1/m;
for t = 1, …, T do
    Fit f_t(x) ∈ ℝ using the current weights;
    Set Z_t ← Σ_{i=1}^{m} w_i exp(−f_t(x_i) y_i);
    Update w_i ← w_i exp(−f_t(x_i) y_i) / Z_t;          // normalize
return F(x) = Σ_{t=1}^{T} f_t(x);
```

1.1 Two trajectories side by side

Minimum fuel



A smoother curve uses less fuel.

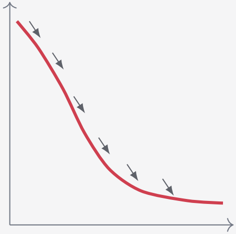
SAME DYNAMICS

Left: minimize fuel.

Right: minimize time.

The middle column names the tradeoff; the figures do the rest.

Minimum time



A steeper curve reaches the target sooner.

1.1 Frame-local background images

Full canvas with readable text

Use a photo, diagram, texture, or generated image on one frame without setting a deck-wide background.

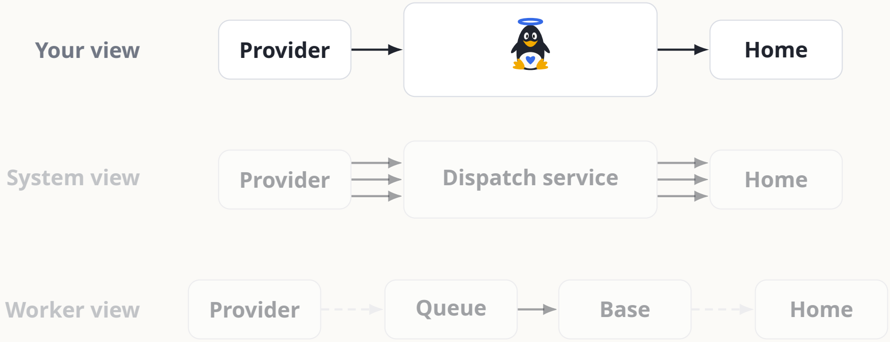
Set the wash color and opacity for contrast. Use local colors when the frame needs light text.

Frame syntax

```
\begin{frame}[
  bgimage=photo.jpg,
  bgoverlay=cookieInk,
  bgoverlayopacity=.56
]
```

1.1 Show one layer at a time

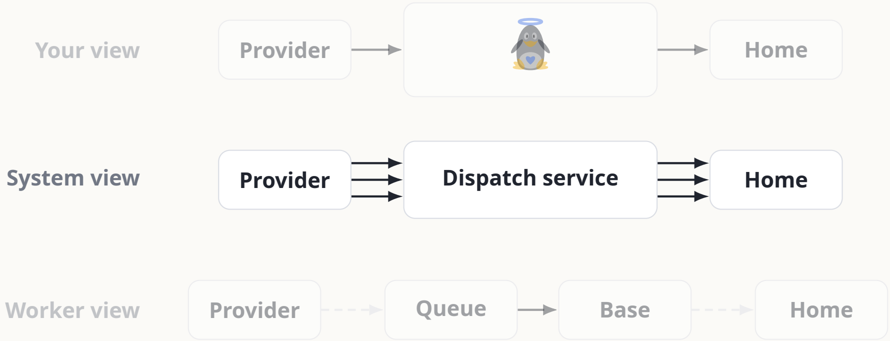
The geometry stays fixed across overlays



Start with the audience’s view of the system.

1.1 Show one layer at a time

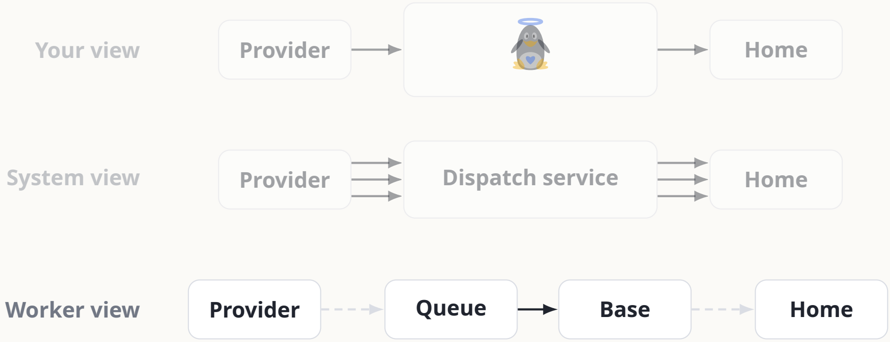
The geometry stays fixed across overlays



Then show the service boundary.

1.1 Show one layer at a time

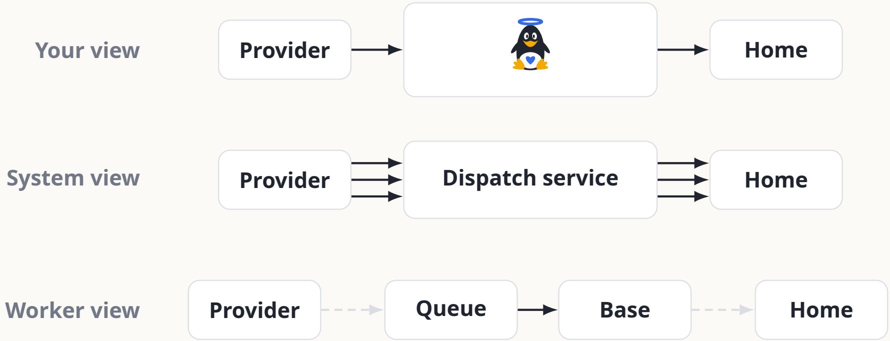
The geometry stays fixed across overlays



Add worker detail when it helps explain the failure mode.

1.1 Show one layer at a time

The geometry stays fixed across overlays



End with all rows visible so the views can be compared.

1.2

SECTION 1 / DESIGN SYSTEM

Overlays

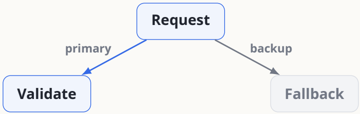
2 / 2

1.2 Reveal without moving the layout

Request

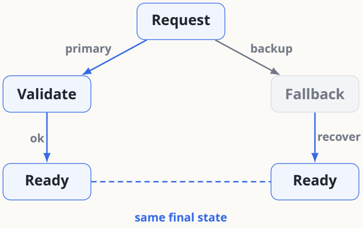
- Reserve the canvas. The request node anchors every overlay.

1.2 Reveal without moving the layout



- Reserve the canvas. The request node anchors every overlay.
- Reveal branches. The primary path carries the accent; fallback stays muted.

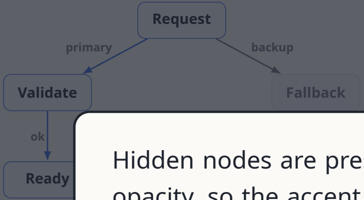
1.2 Reveal without moving the layout



- Reserve the canvas. The request node anchors every overlay.
- Reveal branches. The primary path carries the accent; fallback stays muted.
- Show convergence. Both paths end in the same user-visible state.

Primary	request → validate → ready
Fallback	request → fallback → ready

1.2 Reveal without moving the layout



- Reserve the canvas. The request node anchors every overlay.
- ches. The primary path carries the

ame

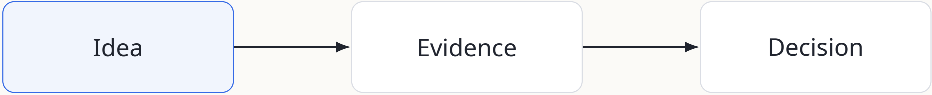
Overlay rule

Hidden nodes are present from the first overlay. Each step changes opacity, so the accent path can appear without moving the text.

Primary	request → validate → ready
Fallback	request → fallback → ready

2
Content patterns

2 Let the footer carry progress



The footer shows location and frame count; the slide can stay on the current step.

Keep tables plain

Feature	None	Simple	Progress
Section page	✔	✔	✔
Subsection page	✔	✔	✔
Progress bar	head	title	foot
Frame numbering	none	count	fraction

Use `booktabs` for rules and `tabularx` for width. Cookie only sets spacing and color.

A larger table

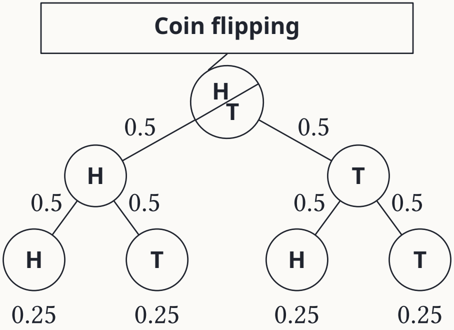
Booktabs rules and aligned numeric columns

Faculty	With $\TeX$	Total	%
Informatics	1 716	2 904	59.09
Science	786	5 275	14.90
Economics	64	4 591	1.39
Arts	69	10 000	0.69
Medicine	8	2 014	0.40
Law	15	4 824	0.31
Education	19	8 219	0.23
Social studies	12	5 599	0.21
Sports studies	3	2 062	0.15

Example data adapted from a university thesis-count table.

A probability tree

TikZ diagram adapted from the source example



One idea per slide.

Use quotes sparingly

> *A good slide should make the argument easier to see.*
>
> — Cookie design principle

A quote works best as a turn in the talk. Don’t use too many!

Speaker notes live in the source

Use `notes=second` for presenter mode

Inside any frame

```
The audience sees this.

\note{
  Remember why this slide matters.
  Add timing, transitions, and
    questions here.
}
```

Output modes

`notes=hide`	slides only
`notes=show`	slides plus notes
`notes=second`	presenter layout
`notes=only`	notes pages only

native Beamer **styled notes page**

3
Configuration

3 Options in one place

Layout

`progressbar`	none, head, frametitle, foot
`numbering`	none, counter, fraction
`numberrail`	none, section, subsection
`covered`	invisible, transparent
`notes`	hide, show, second, only
`toc`	none, aftertitle

Appearance

`accent`	any xcolor name
`background`	light, dark
`block`	transparent, fill
`numberrailcolor`	muted, accent
`titleformat`	regular, smallcaps, allcaps

Compatibility aliases such as `secslide`, `subsecslide`, `notoc`, and `coloraccent` make older awesome-beamer decks easier to move.

Two background commands

Title-page wedge

`\cookietitleimage{image.jpg}`

Clips the image into Cookie's asymmetric title art.

Ordinary frames

`bgimage=image.jpg`

Add the frame keys you need:

```
bgfit=cover
bgoverlay=cookieInk
bgoverlayopacity=.56
```

Global background

`\cookiebackgroundimage{image.jpg}` applies an image to later frames.
`\cookieclearbackgroundimage` removes it.

Build the demo with LuaLaTeX

LOCAL BUILD

Repository command

```
latexmk demo.tex
```

- The local `.latexmkrc` selects LuaLaTeX and Biber for this demo.
- `latexmk` reruns TeX when section, frame, QR, or bibliography data changes.
- If you copy only the theme file into another deck, build that deck with LuaLaTeX.

4

References

4 Citations use normal BibLaTeX

No theme-specific bibliography syntax

Cookie is still ordinary Beamer: the typesetting stack builds on TEX (Knuth 1984), LATEX (Lamport 1994), and Beamer (Tantau et al. 2025).

The design borrows progress bars from Metropolis (Vogelgesang 2015) and the numbered slide grammar of awesome-beamer (Pietzschmann 2024).

Source setup

```
\usepackage[backend=biber,style=authoryear]{biblatex}
\addbibresource{refs.bib}
```

References I

- Knuth, Donald E. (1984). *The TeXbook*. Addison-Wesley.
- Lamport, Leslie (1994). *LaTeX: A Document Preparation System*. 2nd ed. Addison-Wesley.
- Pietzschmann, Lukas (2024). *Awesome-Beamer*. URL: `https://github.com/LukasPietzschmann/awesome-beamer` (visited on 06/21/2026).
- Tantau, Till, Joseph Wright, and Vedran Miletić (2025). *The Beamer Class: User Guide for Version 3.76*. The LaTeX Project. URL: `https://ctan.org/pkg/beamer`.
- Vogelgesang, Matthias (2015). *Metropolis: A Modern Beamer Theme*. URL: `https://github.com/matze/mtheme` (visited on 06/21/2026).

Thank you

Questions, forks, and bug reports are welcome.

Replace with your project URL

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